

CSE - 641 Computer Vision

Weekly Project Report – 7



**Ahmedabad
University**

Ahmedabad University

Genotype-to-Phenotype Modeling of Alzheimer's Disease Using MRI Radiomics and Polygenic Risk Scores

Professor: Dr. Mehul Raval
Teaching Assistant: Shalvi Modi

Team Members

Name	Enrollment Number
Dhriti Gandhi	AU2340030
Ura Modi	AU2340031
Devashree Parikh	AU2340067
Yash Shah	AU2340078

Abstract—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of thinking ability. Studies indicate that genetic factors as well as changes in brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between inheritable genetic risk and structural brain change is not fully understood.

This project examines the association between genetic vulnerability and brain structure using Polygenic Risk Scores (PRS) and MRI-based radiomic features. Important brain regions including the hippocampus and entorhinal cortex are analyzed using MRI scans from the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset.

PRS values are computed to estimate inherited genetic risk, and machine learning models are used to identify relationships between genetic risk and MRI features. The goal is to better understand the role of genetic factors in early brain changes related to Alzheimer disease, potentially before clinical symptoms appear.

I. INTRODUCTION

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk. There have been many research studies that attempted to improve early detection of Alzheimer disease with use of brain imaging methods. Structural magnetic resonance imaging is one of the methods that have been used majorly as it is capable of showing the changes that take place in the brain regions which are affected by disease. One of the affected regions is the hippocampus which is part of the brain that is affected in the Alzheimer disease as it is the part associated with memory. Previous studies were interested in the size of the brain regions. However with the use of the radiomics method, more quantitative features of the MRI images may be obtained that may potentially reveal more details. The authors of the paper [5] would like to create a model of radiomics using MRI images of the hippocampus of the brain to aid in the identification of patients suffering from Alzheimer disease. This would be achieved by the use of data from various cohorts to ensure the reliability of the model. Radiomic features would be achieved by the use of MRI images of the hippocampus. This would then be used to create a model that would aid in the differentiation of patients suffering from Alzheimer disease and healthy patients. This would be achieved by the use of MRI images and the expression of genes present in the blood.

This would then attempt to establish a connection between the genetic features and the biological processes aiming to have a better understanding of the changes taking place in the brain as shown by MRI images.

Recent studies have explored radiomics-based analysis of MRI scans to extract quantitative features that may reveal subtle changes not visible through conventional measurements. For example, Xia et al. [5] developed a hippocampal radiomics model using multi-cohort MRI data to differentiate Alzheimer patients from healthy controls and to investigate biological mechanisms linked to imaging features.

Combining radiomic features with genetic information may help improve early detection and understanding of disease progression. The summary of the selected parameters from the ADNI database is given in the Table 1.

II. DATASET DESCRIPTION

The data used in this project is obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) a multi-centered longitudinal large scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression. This data was acquired following strict protocol guidelines to ensure consistency and reproducibility.

Data in this study has been taken from both ADNI2 and ADNI3 phases. Both phases involve longitudinal data that involves several repeat examinations per patient. Therefore in contrast to a cross-sectional design where only a single instance is considered, a longitudinal design is used here which involves multiple instances per subject.

The study participants are categorized into these primary groups:

- Cognitively Normal (CN)
- Mild Cognitive Impairment (MCI)
- Alzheimer's Disease (AD)

Data available through ADNI include:

- Structural MRI
- PET imaging (FDG-PET and Amyloid PET)
- Cerebrospinal fluid biomarker levels
- Genomic data (APOE genotype)
- Neuropsychological test results (MMSE, CDR, ADAS-Cog)

The filtering of the dataset depends on ADNI2 and ADNI3 phases while the visits used are various these include baseline and follow up visits that are spread throughout different periods of time to analyze disease progression.

The following criteria are taken into account in the selection of the images in this study, Image Description-MPRAGE, Weighting-T1 weighted MRI, Acquisition-3D, Field Strength-3T.

The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability. Due to its comprehensive and longitudinal nature the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction and biomarker discovery in Alzheimer's disease studies.

III. LITERATURE REVIEW

A. Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression

The authors in this paper show how a longitudinal MRI-based paradigm can be utilized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a twostage deep learning model on the T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-conversion, rather than just binary classification.

B. Deep Learning-Based Feature Extraction with MRI Data in Neuroimaging Genetics for Alzheimer's Disease

Another interesting research direction in imaging genetics involves the use of deep learning in order to generate biologically informative representations for use in genetic association studies. With regard to such efforts, Chakraborty et al [4] have introduced an algorithm that incorporates CNNs with GWAS analysis for studying Alzheimer's disease (AD). The major difference between this approach and conventional GWAS methodologies based on ROI-derived features is that in their work, CNNs are used to extract data-driven features from whole 3D MRI volumes; hence, no spatial information is lost. Furthermore, as the name suggests, such features reflect complex patterns of brain atrophy and can be considered as endophenotypes – intermediate quantitative traits that are more biologically representative than actual disease outcomes. Finally, the learned features can be used in GWAS analysis by relating the associated SNPs to each individual; the resulting methodology allows detecting genetic variations that impact brain structure, recovering previously known AD-associated SNPs as well as identifying novel associations. The combination of deep learning algorithms and statistical genetics is clearly advantageous, although this particular framework does not consider multimodal interactions explicitly and does not involve predictive modeling.

IV. METHODOLOGY

As shown in Fig 1. this MRI-based methodology for Alzheimer's Disease integrates genetic risk factors to enable multi-modal modeling of disease progression.

As in Phase One, the MRI imaging process, including preprocessing and segmentation, is used to extract structural features from brain scans. Specifically, tools such as FastSurfer

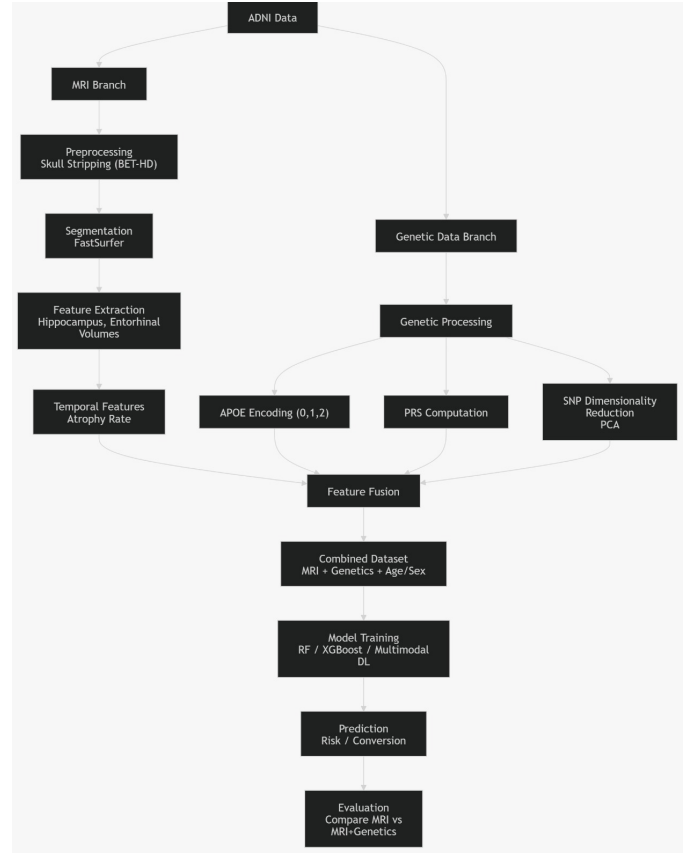


Fig. 1: Descriptive Flowchart for methodology

are employed to obtain measurements of key regions, including hippocampal and entorhinal cortex volumes. In addition, atrophy rates are computed using temporal features derived from longitudinal imaging data.

To achieve a consistent biological representation of genomic features across different subjects from ADNI Phases 2 and 3, an integrated genetic data processing pipeline was developed. Due to the use of different genotyping platforms across phases, the available single nucleotide polymorphisms (SNPs) varied between subjects, resulting in inconsistent feature sets that hindered downstream machine learning model construction. To address this issue, a common set of SNPs shared between both phases was identified. This set was further aligned with SNPs present in the genome-wide association study (GWAS) summary statistics used for polygenic risk score (PRS) computation. Subsequently, standard quality control (QC) procedures were applied using PLINK. These included SNP-level filtering based on missing genotype rates (`--geno`), individual-level filtering based on missingness (`--mind`), and filtering based on minor allele frequency (`--maf`). Following QC, linkage disequilibrium (LD) pruning was performed to remove highly correlated and redundant SNPs, thereby reducing dimensionality and ensuring a more informative and independent SNP feature space for downstream analysis.

After preprocessing, polygenic risk scores (PRS) were calculated for each participant using effect sizes derived from

TABLE I: Summary of Selected ADNI Dataset Parameters

Parameter	Selection
ADNI Phases	ADNI-2 and ADNI-3
Imaging Modality	Structural MRI
MRI Sequence	T1-weighted MPRAGE
Field Strength	3 Tesla (3T)
Visit Type	Temporal
Subject Groups	CN, MCI, AD
Image Format	NIfTI (.nii)
Genetic Data	APOE Genotype

genome-wide association studies (GWAS), providing an estimate of the cumulative genetic risk for Alzheimer’s Disease. In parallel, the filtered SNP data was numerically encoded and standardized prior to applying principal component analysis (PCA) to obtain a lower-dimensional representation. The resulting principal components capture the underlying genetic variability across individuals and can be used to account for population stratification. Finally, the computed PRS values and SNP-derived principal components were combined to form a unified genomic feature representation. This combined feature set can be directly integrated with MRI-derived features and clinical variables to evaluate the predictive potential of genetic factors in Alzheimer’s Disease onset and progression.

The MRI-derived features and genetic information are then integrated into a unified dataset. This dataset also incorporates demographic variables such as age and sex. Subsequently, machine learning or deep learning models are applied to predict Alzheimer’s Disease risk or progression, for example, the conversion from Mild Cognitive Impairment (MCI) to Alzheimer’s Disease (AD).

The multi-modal approach provides a comprehensive framework that captures both structural brain changes (phenotype) and inherited genetic risk (genotype). This integration improves predictive accuracy and enables earlier detection compared to models that rely solely on MRI-based features.

V. WORK COMPLETED IN WEEK-7

This week concentrated on the development of the genetic data processing pipeline for ADNI Phases 2 and 3. A set of SNPs common to both phases was found and matched with the summary statistics from the GWAS for calculating the PRS. Quality control procedures using PLINK were performed, after which LD pruning was conducted.

Further on, the polygenic risk score (PRS) was calculated to evaluate the genetic risk, while PCA was performed with normalized SNP data to account for genetic diversity. At the same time, MRIs data pre-processing was done via the HD-BET base code for skull stripping, followed by segmentation as shown in Fig. 2.

As the final step, the PRS values along with the principal components of the SNP data were integrated into one genomic dataset.

VI. PLAN FOR WEEK-8

For week 8, the main goal is to implement a neural network approach for feature extraction in MRI scans, coupled with

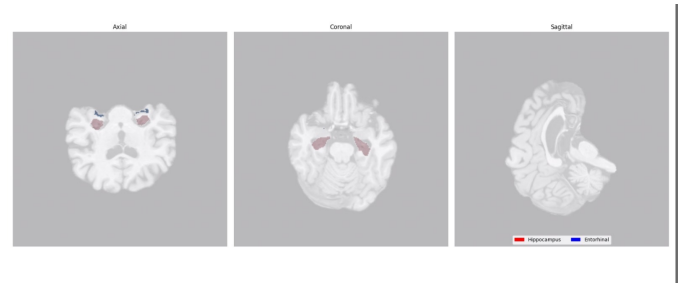


Fig. 2: Segmented 3D brain MRI highlighting different anatomical regions across multiple views

genomic data. This would enable the design of a model that can extract useful representations from imaging data and subsequently relate them with genetic markers such as PRS and SNP-derived PCs. The overall idea here is to analyze the association between alterations in brain structure and genetic risks associated with Alzheimer’s disease. Another aspect that will be considered is how to appropriately merge multiple sources of data.

REFERENCES

- [1] E. C. Bellenguez et al., “New insights into the genetic etiology of Alzheimer’s disease,” *Nature Genetics*, vol. 54, pp. 412–436, 2022.
- [2] C. Leonenko et al., “Polygenic risk and hazard scores for Alzheimer’s disease prediction,” *Alzheimer’s & Dementia*, vol. 17, no. 2, pp. 294–306, 2021.
- [3] J. Li et al., “Imaging genetics in Alzheimer’s disease,” *Frontiers in Neuroscience*, vol. 15, 2021.
- [4] D. Chakraborty, Z. Zhuang, H. Xue, M. B. Fiecas, X. Shen, and W. Pan, “Deep Learning-Based Feature Extraction with MRI Data in Neuroimaging Genetics for Alzheimer’s Disease,” *Genes*, vol. 14, no. 3, p. 626, Mar. 2023, doi: 10.3390/genes14030626.
- [5] H. Xia et al., “A multi-cohort hippocampal radiomics model in Alzheimer’s disease,” *Translational Psychiatry*, 2024.
- [6] A. Biffi, C. D. Anderson, R. S. Desikan, M. Sabuncu, L. Cortellini, N. Schmansky, D. Salat, J. Rosand, et al., “Genetic variation and neuroimaging measures in Alzheimer disease,” *Archives of Neurology*, vol. 67, no. 6, pp. 677–685, 2010, doi: 10.1001/archneurol.2010.108.
- [7] I. de Rojas, S. Moreno-Grau, N. Tesi, et al., “Common variants in Alzheimer’s disease and risk stratification by polygenic risk scores,” *Nature Communications*, vol. 12, p. 3417, 2021, doi: 10.1038/s41467-021-22491-8.